Number theory – 6

Arithmetic functions and Dirichlet convolution

Homework

14 сентября 2026 года

Theoretical minimum. An arithmetic function is multiplicative if $f(mn) = f(m)f(n)$ for coprime arguments and $f(1) = 1$; the basic ones are φ, d, σ, μ and $\mathbf{1}$. Under the Dirichlet convolution $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$ they form a commutative ring with identity $[n = 1]$, invertible elements are those with $f(1) \neq 0$, and the convolution of multiplicative functions is multiplicative. Since $\mu * \mathbf{1} = \varepsilon$, one has Möbius inversion: $g = f * \mathbf{1}$ if and only if $f = g * \mu$. Standard identities: $\mathbf{1} * \mathbf{1} = d$, $\mathrm{id} * \mathbf{1} = \sigma$, $\varphi * \mathbf{1} = \mathrm{id}$, whence $\varphi(n) = n \prod_{p|n}(1 - 1/p)$. Dirichlet series turn convolution into multiplication, giving $\sum \mu(n)n^{-s} = 1/\zeta(s)$, $\sum d(n)n^{-s} = \zeta(s)^2$, $\sum \varphi(n)n^{-s} = \zeta(s-1)/\zeta(s)$; convergence and summation belong to Series and infinite products – 5 and – 6. Average orders: $\sum_{n \leqslant x} d(n) \sim x \ln x$, $\sum_{n \leqslant x} \varphi(n) \sim \frac{3}{\pi^2} x^2$.

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1. Prove that $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$, that $\sum_{d|n} \frac{\mu(d)}{d} = \frac{\varphi(n)}{n}$ and that $\sum_{d|n} \mu(d) \ln d = -\Lambda(n)$, where Λ is the von Mangoldt function; then express Λ as a convolution and read off the Dirichlet series $-\zeta'(s)/\zeta(s)$.

2. Prove that the number of monic irreducible polynomials of degree n over $\mathbb{F}_q$ equals $\frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}$, and deduce that this number is positive for every n and q. (The finite fields themselves are Polynomials – 6.)

3. Prove that the number of necklaces of n beads in k colours up to rotation equals $\frac{1}{n} \sum_{d|n} \varphi(d) k^{n/d}$, first by Burnside's lemma (Abstract algebra – 4) and then by Möbius inversion applied to the count of aperiodic words.

4. Prove that $d(n) = O(n^{\varepsilon})$ for every $\varepsilon > 0$ while $\limsup \frac{\ln d(n) \, \ln \ln n}{\ln n} = \ln 2$, and determine the maximal order of $\sigma(n)/n$.

5. Prove that $\sum_{n\leqslant x}\mu(n)/n$ is bounded, and explain what its tending to zero is equivalent to. Then prove the elementary identity $\sum_{n\leqslant x}\lfloor x/n\rfloor = \sum_{n\leqslant x} d(n)$ and use it for a second proof of the average order of d.

6. Let f be multiplicative. Prove that $f(n)\to 0$ as $n\to\infty$ through the integers if and only if $f(p^k)\to 0$ as the prime power $p^k\to\infty$, and classify the completely multiplicative f with $|f(n)|=1$ for all n.

7. For a non-negative integer n define $\operatorname{rad}(n)=1$ for $n\in\{0,1\}$ and $\operatorname{rad}(n)=p_1p_2\cdots p_k$, the product of the distinct primes of n, otherwise. Find all polynomials f with non-negative integer coefficients such that $\operatorname{rad}(f(n))$ divides $\operatorname{rad}(f(n^{\operatorname{rad}(n)}))$ for every non-negative integer n. (IMO shortlist 2012, N5)

8. Prove that $\sum_{n\leqslant x}\frac{\varphi(n)}{n}=\frac{6}{\pi^2}x+O(\ln x)$ and that the number of Farey fractions of order N is $\frac{3}{\pi^2}N^2+O(N\ln N)$; relate the two computations.

9. Prove that $\sigma(n)\varphi(n)<n^2<\sigma(n)\varphi(n)\cdot\frac{\pi^2}{6}$ for all $n>1$, and determine for which n the ratio $n^2/(\sigma(n)\varphi(n))$ is closest to its extremes.

10. Let ν be a positive irrational number and m a positive integer. A pair (a,b) of positive integers is *good* if $a\lceil b\nu\rceil - b\lfloor a\nu\rfloor = m$, and a good pair (a,b) is *excellent* if neither of the pairs $(a-b,b)$ and $(a,b-a)$ is good. Prove that there are finitely many excellent pairs, and determine their number. (IMO shortlist 2013, N7)